



Indian Institute of Technology Kharagpur

EXAMINATION ANSWER SCRIPT

Stamp/Signature of the Invigilator

EXAMINATION: Mid Semester

SEMESTER: Autumn 2024-25

Roll Number										Section		Name	
Subject Number	C	S	3	1	0	0	7			Subject Name	Computer Organization and Architecture		
Department/Centre/School												Additional Sheets	

Important Instructions and Guidelines for Students

1. You must occupy your seat as per the Examination Schedule/Sitting Plan.
2. Do not keep mobile phones or any similar electronic gadgets with you even in the switched off mode.
3. Loose papers, class notes, books or any such materials must not be in your possession; even if they are irrelevant to the subject you are taking examination.
4. Data book, codes, graph papers, relevant standard tables/charts or any other materials are allowed only when instructed by the paper-setter.
5. Use of instrument box, pencil box and non-programmable calculator is allowed during the examination. However, the exchange of these items or any other papers (including question papers) is not permitted.
6. Write on both sides of the answer-script and do not tear off any page. **Use last page(s) of the answer-script for rough work.** Report to the invigilator if the answer-script has torn or distorted page(s).
7. It is your responsibility to ensure that you have signed the Attendance Sheet. Keep your Admit Card/Identity Card on the desk for checking by the invigilator.
8. You may leave the Examination Hall for wash room or for drinking water for a very short period. Record your absence from the Examination Hall in the register provided. Smoking and the consumption of any kind of beverages are strictly prohibited inside the Examination Hall.
9. Do not leave the Examination Hall without submitting your answer-script to the invigilator. **In any case, you are not allowed to take away the answer-script with you.** After the completion of the examination, do not leave your seat until the invigilators collect all the answer-scripts.
10. During the examination, either inside or outside the Examination Hall, gathering information from any kind of sources or exchanging information with others or any such attempt will be treated as 'unfair means'. Don't adopt unfair means and also don't indulge in unseemly behavior.

Violation of any of the above instructions may lead to severe punishment.

Signature of the Student

To be Filled by the Examiner

Question Number	1	2	3	4	5	6	7	8	9	10	Total
Marks Obtained											
Marks Obtained (in words)				Signature of the Examiner				Signature of the Scrutineer			



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Mid-Autumn Semester Examination 2024-25

Date of Examination: 19-Sept-2024

Session: AN

Duration: 2 hrs.

Full Marks: 60

Subject No.: CS31007

Subject: Computer Organization and Architecture

Department: Computer Science and Engineering

Specific charts, graph paper, log book etc., required: NONE

Special Instructions (if any): No clarification will be provided during the exam. State any assumptions made.

Answers must be written only in the boxes provided in the booklet.

Rough works must be done only in the specified spaces provided in the booklet.

ANSWER ALL THE QUESTIONS

SPACE FOR ROUGH WORK

Q1. Answer the following.

[5 x 4 = 20]

- a) Clearly explain the differences between *indirect addressing* and *indexed addressing* with the help of examples.

- b) Mention *four differences* between RISC and CISC architectures.

- c) Consider a RISC architecture with 32-bit registers and 32-bit instructions. There are 75 distinct instruction types, and 16 registers (R0, R1, ..., R15). Show possible instruction encodings for R-type and I-type instructions (R-type and I-type instructions are defined as in MIPS32).

- d) Consider a control unit generating the control signals for a data path. These control signals are divided into five *mutually exclusive groups* as shown below:

Groups	G1	G2	G3	G4	G5
Control Signals	3	6	12	14	2

How many bits are saved to encode a control word using *diagonal micro-instruction encoding*, as compared to *horizontal micro-instruction encoding*?

- e) What are pseudo-instructions in MIPS32 assembly language (as in QtSpim simulator)? Show how the following pseudo-instructions are translated into normal MIPS32 instructions?

- i) `li        0x2345ABCD`
- ii) `move     R1 ,R2`

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Q2. Answer the following.

[4 + 6 = 10]

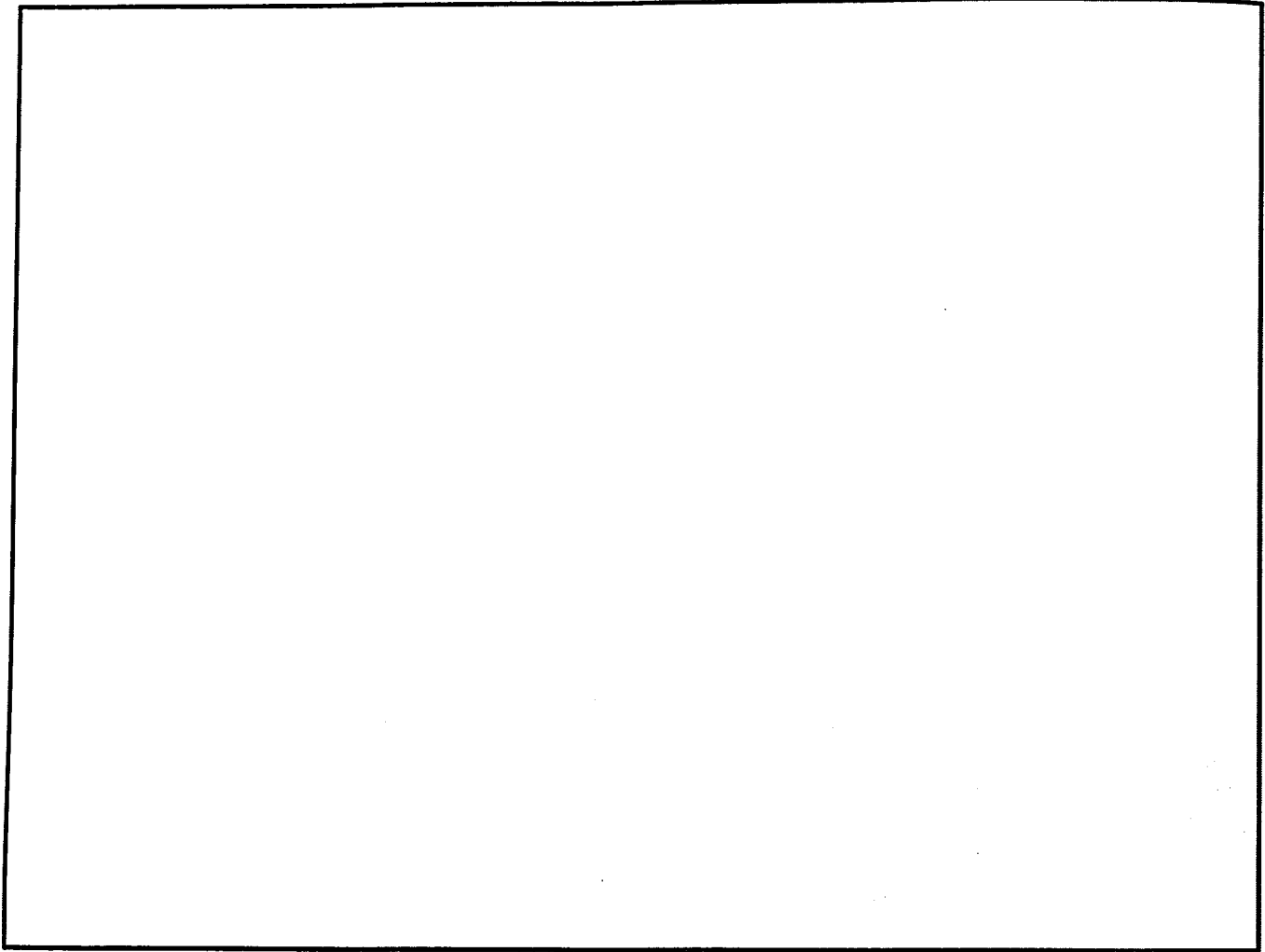
- a) Consider the evaluation of the expression $F = X + (Y - Z)$, where X , Y and Z are integer operands stored in memory, and F is the result also to be stored in memory. Write assembly language code segments to evaluate the expression for the following instruction set architectures. Make relevant assumptions as necessary.

- i) Stack architecture
- ii) Accumulator-based single address architecture
- iii) Two address architecture
- iv) Load-Store architecture (similar to MIPS32)

- b) Suppose we are running a set of programs on a RISC processor, for which the following instruction mix is observed:

Instruction Type	Frequency	Cycles per Instruction (CPI)
LOAD	20 %	5
STORE	10 %	3
Integer ALU	40 %	1
Floating-point ALU	15 %	3
Branch	15 %	2

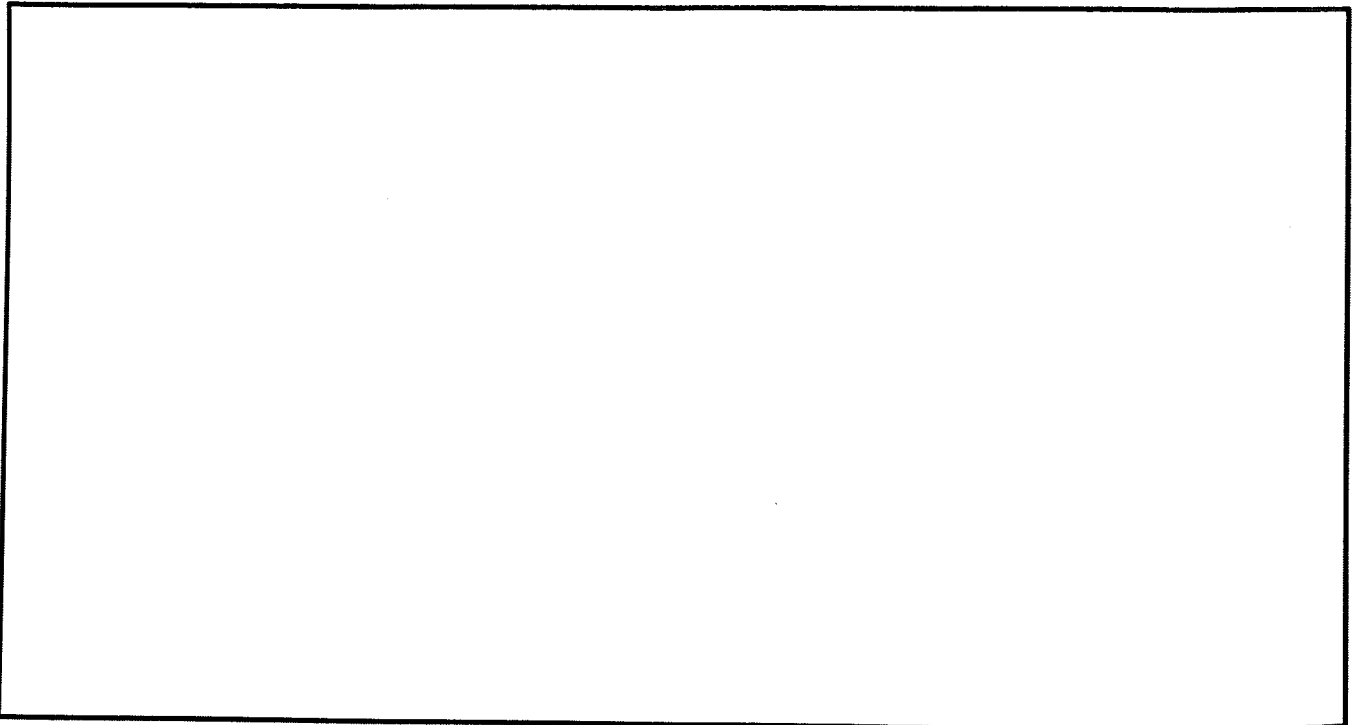
We carry out a design enhancement by which the CPI of LOAD instructions reduces from 5 to 3, and that for Floating-point ALU instructions from 3 to 2. Using Amdahl's law, estimate the overall performance improvement.



Q3. Answer the following.

[2 + 6 + 2 = 10]

- a) Draw schematic diagrams for a 6-transistor SRAM cell. Briefly explain how data read operation is carried out in the cell.



- b) Consider a processor with 24-bit address bus and 8-bit data bus. It is required to build a 16 Mbyte (16M x 8) memory system using 2 Mbyte (2M x 8) memory chips as basic building blocks.
- What are the typical pins that a 2 Mbyte memory chip will have?
 - Draw a detailed schematic diagram showing the CPU-memory interfacing for building the 16 Mbyte memory system. *Do not use memory interleaving.*
 - Mention the range of addresses (*in hexadecimal*) that will get mapped to each of the memory chips.

- c) Briefly state how the concept of *memory interleaving* helps in improving the processor-memory bandwidth?

Q4. Answer the following.

[4 + 3 + 3 = 10]

- a) Consider a 2-level memory hierarchy with separate instruction cache (I-cache) and data cache (D-cache) in Level-1, and main memory in Level-2. Assume that the clock frequency is 1 GHz, the miss penalty is 25 clock cycles (for both read and write), 98% of instruction references are found in I-cache, 90% of data references are found in D-cache, 15% of the total memory accesses are for data, and the cache access time (including hit time) is 1 clock cycle. Determine the average access time of the memory system in nanoseconds.

- b) Briefly explain how *block placement* and *block identification* take place in (i) a direct-mapped cache, (ii) a set-associative cache.

- c) A 256 Kbyte ($256K \times 8$) four-way set-associative cache is byte-addressable and the cache block size is 64 bytes. Assume that the CPU generates a 24-bit address.
- i) How wide are the TAGs in the cache?
 - ii) How many TAGs need to be compared to detect a cache hit?
 - iii) To which SET will the following memory addresses (specified in hexadecimal) be mapped: 0x 02A0C6 and 0x 028DA7 ?

Q5. Answer the following.

[4 + (3 + 3) = 10]

- a) Assume that instruction execution in an implementation of the MIPS32 processor is carried out in five steps: IF, ID, EX, MEM and WB. Write down the micro-instructions for executing the following instructions. *Do not show the control signals.*

- i) LOAD R5, 75(R8)
- ii) BEQZ R1, Label

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- b) Suppose we want to design the hardware for calculating the GCD of two numbers **A** and **B** using the algorithm specified in the following pseudo-code:

```
function GCD (int A, int B)
{
    while (A != B) {
        if (A < B) B = B - A;
        else A = A - B;
    }
    return A;
}
```

- i) Draw the data path for the hardware required for the computation, and clearly identify the control signals in the diagram.

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- ii) Draw a flowchart specifying the control signals to be activated in the various *states* during the computation.

